Edge–Cloud Reinforcement Learning for Transferable HVAC Control Across Vietnamese Climate Zones

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Abstract

Heating, ventilation, and air-conditioning (HVAC) systems account for a large fraction of commercial-building energy use in Vietnam, where rapid urbanisation and high cooling demand across a long north–south climate gradient make HVAC control a priority for both operators and policy makers. Reinforcement learning (RL) is a natural candidate for adaptive HVAC control, but most published evaluations train and test policies on a single climate in offline simulation, leaving three practical questions open: how a trained policy behaves under tight inference-latency budgets, how quickly it loses accuracy under climate-driven distribution shift, and how its retraining lifecycle should be organised. We address these questions with an edge–cloud adaptive RL framework evaluated on a Vietnam multi-city protocol covering three ASHRAE climate zones: Ho Chi Minh City and Can Tho (zone 0A, extremely hot–humid), Da Nang (zone 1A, hot–humid), and Hanoi (zone 2A, warm–humid). Commercial-building archetypes from the HOT benchmark are paired with city-specific TMYx weather files to obtain a reproducible multi-climate testbed. The edge layer runs the policy locally with $p_{95} < 0.002$ ms latency and validates each action against safety constraints; the cloud layer performs multi-context Proximal Policy Optimisation (PPO) retraining and pushes versioned policy updates that must pass a deployment-aware acceptance test. The baseline analysis recovers the expected north-to-south cooling-energy gradient: HVAC consumption in Ho Chi Minh City is 1.44× that of Hanoi for the OfficeSmall archetype and 1.68× for OfficeMedium. Edge inference is approximately 71,000× faster than cloud inference at the 95th percentile. Pure-PPO policies (single-context, multi-context, and edge-only) cut the comfort-violation rate by 1.45× relative to a static-setpoint baseline (mean across the ten Vietnam contexts; 19.8% vs. 28.7%) but consume 58.7% more HVAC energy, exposing an energy–comfort trade-off that PPO does not resolve on its own. The edge–cloud adaptive controller closes this gap: it matches the comfort profile of the static baseline (violation rate 26.7% vs. 28.7%) and reaches a worst-context energy regret of 640.8 kWh, close to the static baseline (644.7 kWh) and far below the single-context PPO family (3,336.2 kWh).

Keywords: Reinforcement learning; HVAC control; Edge–cloud architecture; Transfer learning; Vietnam; Tropical buildings; Climate-zone generalisation

1 Introduction

Vietnam's commercial-building stock is expanding at a pace that places HVAC operation squarely at the centre of national energy and climate policy. Cooling loads dominate operation throughout the south, where annual cooling-degree days exceed 3,800, while Hanoi exhibits a seasonally mixed regime with warm–humid summers and cooler winters. In 2022 alone, Vietnam's cooling sector emitted an estimated 64.68 million tonnes of CO_2-equivalent, with residential and commercial air-conditioning responsible for over one third of that total [1]. Against this backdrop, intelligent HVAC control is among the highest-leverage mitigation strategies available at the building stock level.

Reinforcement learning is an attractive candidate for adaptive HVAC control because it can optimise long-horizon energy–comfort trade-offs without requiring an analytically tractable building model [2, 3]. Despite a decade of encouraging simulation results, two structural obstacles continue to inhibit operational adoption. First, *latency*: cloud-hosted inference incurs network round-trips that can exceed control-loop deadlines, particularly in fast-responding HVAC zones or during demand-response events. Second, *distribution shift*: a policy trained on one city's climate silently degrades when deployed in another city or season unless its retraining lifecycle is explicitly engineered. Vietnam's $\sim$ 1,700 km north–south climate gradient—traversing ASHRAE zones 0A through 2A within a single jurisdiction—furnishes a compact and operationally relevant testbed for both obstacles.

We propose an edge–cloud adaptive RL framework that decouples latency-critical policy execution from compute-intensive learning (Figure 1). The edge layer executes the deployed policy locally, validates each proposed action against safety constraints, and logs operational trajectories. The cloud (or high-performance computing, HPC) layer aggregates these logs, trains candidate policies on multi-city source contexts, and deploys versioned updates only when a deployment-aware acceptance criterion is satisfied. The framework is evaluated on a Vietnam-specific experimental matrix constructed from HOT commercial-building models [4] paired with city-specific TMYx weather files drawn from the EnergyPlus weather catalogue [5].

The contributions of this paper are as follows:

1. A deployment-aware edge–cloud RL architecture that separates low-latency edge inference from cloud-based multi-city retraining under an explicit, versioned policy lifecycle.

2. A reproducible Vietnam multi-city evaluation protocol that combines HOT commercial-building archetypes with city-specific TMYx weather data across three ASHRAE climate zones.

3. Quantitative evidence that ASHRAE-style rule-based scheduling offers no useful energy–comfort advantage over static setpoint control in tropical climates: it saves energy only by tolerating much higher comfort violation.

4. An empirical characterisation of the energy–comfort trade-off induced by PPO under tropical Vietnamese conditions, showing that the pure-PPO variants reduce comfort violations relative to static control (19.8% vs. 28.7%) but at the price of 58.7% higher HVAC energy. This finding directly motivates the proposed edge–cloud adaptive

controller, which preserves the comfort profile of the static baseline while operating at energy levels comparable to it.

2 Related Work

2.1 Reinforcement Learning for Building Control

Reinforcement learning casts HVAC control as a sequential decision-making problem in which a policy observes thermal and contextual state and emits setpoint actions [2]. Among policy-gradient methods, Proximal Policy Optimisation (PPO) is widely adopted in continuous-control settings owing to its clipped surrogate objective and favourable training stability [6]. Applied to buildings, PPO and related actor–critic methods have repeatedly demonstrated energy savings over rule-based baselines in simulation [7, 3, 8]. The persistent gap between simulation-trained RL and deployable control, however, is not principally one of reward design but of *operationalisation*: the simultaneous management of timing constraints, climate-driven distribution shift, and the policy lifecycle.

2.2 Transfer Learning Across Building and Climate Contexts

Each building–climate pairing constitutes a distinct control context, and transfer learning has been studied as a means of reusing policies across contexts [9, 10]. The HOT dataset [4] provides 159,744 building–weather combinations spanning 19 ASHRAE climate zones, enabling systematic transfer studies at scale. The literature has so far concentrated on building-to-building and weather-to-weather transfer in temperate and cold climates; transfer across the tropical and subtropical gradient that characterises Southeast Asia remains largely unstudied. Vietnam, encompassing zones 0A through 2A within a single rapidly urbanising jurisdiction, is a particularly informative testbed for closing this gap.

2.3 Edge–Cloud Architectures for Cyber-Physical Control

Edge–cloud AI separates low-latency inference from compute-intensive learning [11]. For HVAC control this split offers a natural pathway to reduce online latency while preserving the capacity to retrain policies under distribution shift. Recent work on edge inference has emphasised the latency–accuracy trade-off and the risk of policy staleness when retraining cycles are prolonged [12, 13]. Our framework extends this line of work by introducing a multi-city retraining protocol and a deployment-aware acceptance criterion that jointly account for energy performance and cross-city transfer fidelity.

3 Problem Formulation

3.1 Multi-City HVAC Control as a Contextual MDP

We model HVAC control as a family of Markov decision processes (MDPs) indexed by a city–building context c:

$$\mathcal{M}_c = \langle \mathcal{S},\, \mathcal{A},\, P_c,\, R_c,\, \gamma \rangle,$$

where $\mathcal{S}$ is the shared state space, $\mathcal{A}$ is the shared action space of zone setpoints, P_c is the context-conditioned transition kernel encoding city climate and building thermal dynamics, R_c is the reward, and $\gamma \in (0, 1)$ is the discount factor.

The state vector s_t comprises indoor zone temperature (T_{in}), outdoor dry-bulb temperature (T_{out}), outdoor relative humidity (RH), zone occupancy count, HVAC electricity consumption at the previous step (extracted from the `Electricity:HVAC` EnergyPlus meter), and time-of-day sinusoidal features. The action $a_t = [\text{heat\_sp}, \text{cool\_sp}] \in [16, 24] \times [20, 30]$ °C specifies zone-level dual setpoints applied to EnergyPlus `Schedule:Constant` actuators. The instantaneous reward balances energy consumption, thermal discomfort, and actuation smoothness:

$$r_t = -\big(0.12\, e_t + 2.0\, d_t + 0.05\, u_t\big),$$

where e_t (kWh) denotes HVAC electricity over a 15-min step, d_t (°C) is the summed deviation below 20 °C and above 24 °C relative to the ASHRAE 55 comfort band [14], and $u_t = \|a_t - a_{t-1}\|_1$ penalises action chatter.

3.2 Transfer and Deployment Objective

Let $\mathcal{C}_{\text{src}}$ denote the set of source cities used for policy training and $\mathcal{C}_{\text{tgt}}$ the set of target cities reserved for transfer evaluation. In the present protocol, Ho Chi Minh City and Can Tho form the source pool; Da Nang and Hanoi are held out from training and used only for cross-zone transfer evaluation. Transfer is reported through target/source energy ratios, target-context energy, and worst-context energy regret. Deployment decisions are governed by a fixed acceptance rule that jointly checks energy use, comfort-violation regression, inference latency, and action instability; the rule is used as an operational gate rather than as a separately reported scalar score.

4 Edge–Cloud Adaptive Reinforcement Learning

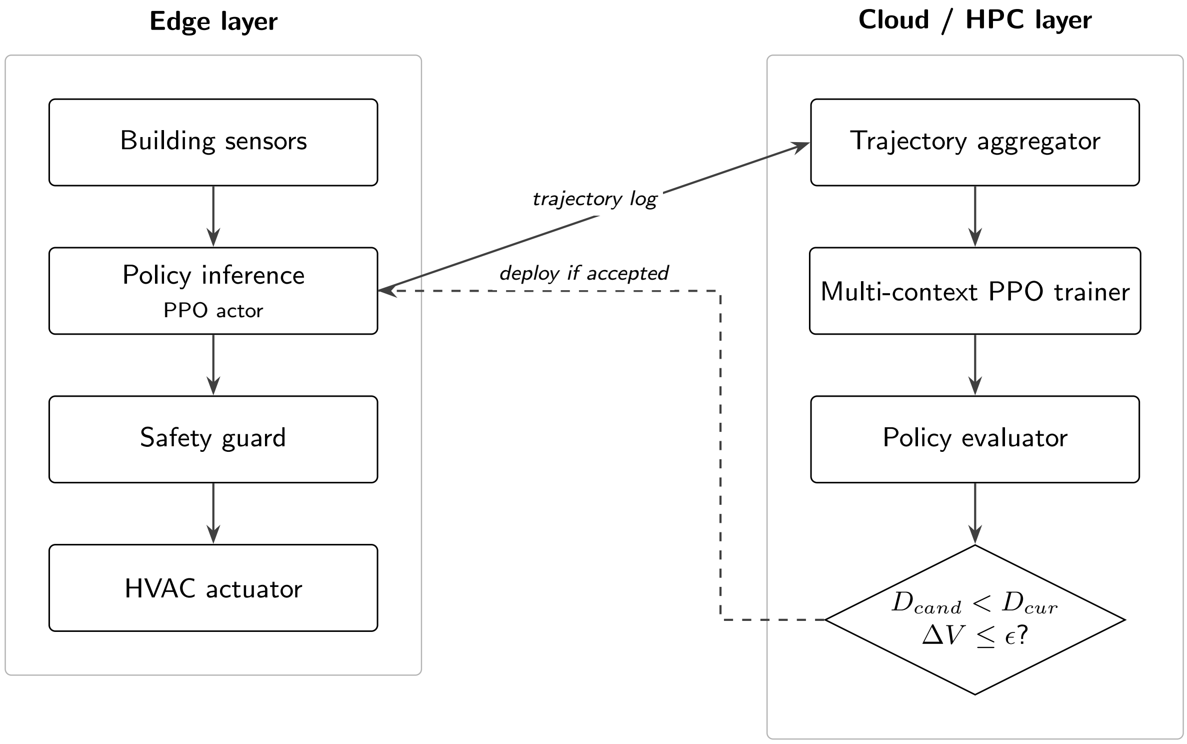

Figure 1: Edge–cloud adaptive RL architecture. Edge devices execute and validate actions; the cloud trains and promotes policy updates.

4.1 Edge Layer

The edge layer executes the 15-minute control loop through six steps:

1. Acquire sensor observations (T_{in}, T_{out}, RH, occupancy, HVAC power, time features).
2. Normalise the state vector using running statistics maintained at the edge.
3. Perform local inference using the deployed policy artifact.
4. Validate the proposed action against hard setpoint bounds ($[16, 24] \times [20, 30]$ °C) and the dual-setpoint separation constraint cool_sp $\geq$ heat_sp $+ 1.5$ °C.
5. Dispatch the validated action to EnergyPlus `Schedule:Constant` actuators through the Python-API callback.
6. Persist state, action, reward components, constraint violations, and a wall-clock timestamp to the trajectory log.

Every deployed policy carries a version tag and a SHA-256 checksum so that rollback is possible if performance degradation is detected after deployment.

4.2 Cloud Layer

The cloud layer ingests trajectory logs from the edge, assembles multi-city training batches from the source contexts (Ho Chi Minh City and Can Tho), trains candidate policies via multi-context PPO using Stable-Baselines3 [15], and evaluates each candidate on a validation split drawn only from the source contexts. A candidate is promoted to deployment only when the fixed acceptance rule improves over the currently deployed policy *and* the comfort-violation regression remains below the allowed threshold $\epsilon_V = 0.02$.

4.3 Policy Update Protocol

The update protocol, summarised in Figure 2, proceeds as follows.

1. The currently deployed edge policy operates for a fixed update interval (default: 96 timesteps, approximately 24 h).
2. The accumulated trajectory log is uploaded to the cloud layer.
3. Candidate policies are trained on the union of logged and simulated source-city contexts.
4. Candidates are evaluated on a held-out validation split from the source contexts.
5. The best candidate is deployed to the edge if and only if the acceptance rule improves and $\Delta V \leq \epsilon_V$; otherwise the incumbent policy is retained.

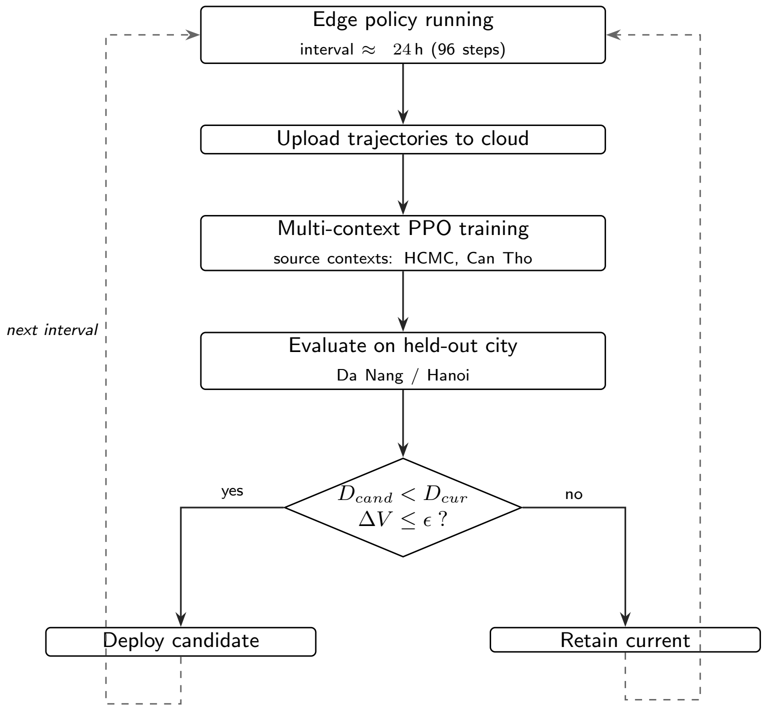

Figure 2: Policy update protocol with deployment-aware acceptance.

5 Experimental Setup

5.1 Vietnam Multi-City Protocol

We use commercial-building archetypes from the HOT dataset [4]—specifically the EnergyPlus 24.1 epJSON models pre-calibrated for ASHRAE zone 0A (Ho Chi Minh City)—paired with city-specific TMYx weather files from climate.onebuilding.org [5]. Holding the building archetype fixed while varying weather isolates the effect of climate-driven distribution shift from that of building-type variation, a standard procedure in multi-climate building simulation [16]. The simulation engine is EnergyPlus 24.1.0 invoked through the official Python API [17]; setpoints are applied through `Schedule:Constant` actuators discovered at runtime. PPO policies are trained with Stable-Baselines3 [15].

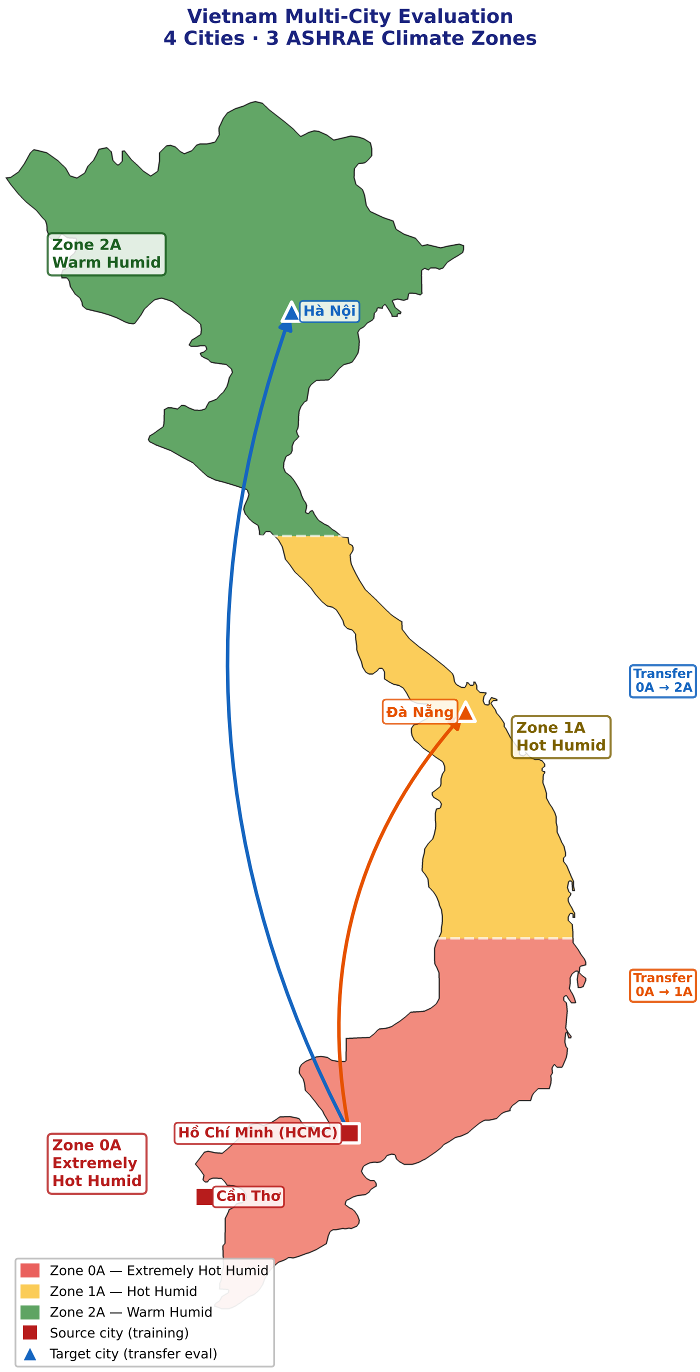

Figure 3: Vietnam city set spanning ASHRAE zones 0A, 1A, and 2A.

Table 1: Cities and ASHRAE climate zones in the Vietnam evaluation protocol.

City	ASHRAE Zone	Description	Role
Ho Chi Minh City	0A	Extremely hot–humid	Source (train)
Can Tho	0A	Extremely hot–humid	Source (train)
Da Nang	1A	Hot–humid	Target (transfer)
Hanoi	2A	Warm–humid	Target (transfer)

5.2 Scenario Matrix

Table 2 defines the five evaluation scenarios illustrated in Figure 4. Because Can Tho is part of the source pool, V1–V2 are treated as same-zone source-domain checks rather than true held-out transfer cases. V3 evaluates a single climate-zone shift (0A $\rightarrow$ 1A); V4–V5 evaluate a two-zone shift (0A $\rightarrow$ 2A) without and with concurrent building-type variation, respectively.

Table 2: Experimental scenario matrix. Ho Chi Minh City and Can Tho provide source training data; Da Nang and Hanoi are held-out transfer targets.

Scenario	Building	Training scope	Evaluation city	Evaluation role
V1	OfficeSmall	Source pool	Can Tho	Seen-source same-zone check
V2	OfficeMedium	Source pool	Can Tho	Seen-source same-zone check
V3	OfficeSmall	Source pool	Da Nang	Held-out climate shift (0A$\rightarrow$1A)
V4	OfficeSmall	Source pool	Hanoi	Held-out climate shift (0A$\rightarrow$2A)
V5	OfficeMedium	Source pool	Hanoi	Held-out building + climate shift

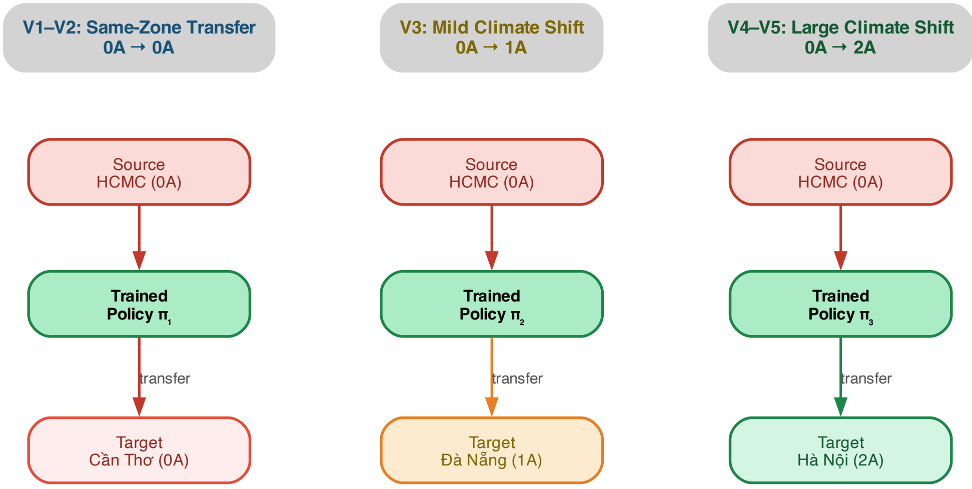

Figure 4: Evaluation scenarios from tropical source cities to held-out target climates, with Can Tho retained as a source-domain same-zone check.

5.3 Compared Methods

1. **Static setpoint:** fixed heating setpoint 20 °C and cooling setpoint 24 °C throughout the episode.
2. **ASHRAE-style rule-based:** outdoor-temperature-conditioned setpoints following the ASHRAE 55 seasonal guidance [14].
3. **Single-context PPO:** a family of PPO checkpoints, each trained on one city–building context; source-trained checkpoints are used for the cross-city transfer analysis.
4. **Multi-context PPO:** PPO trained jointly on both source cities and on all source building types via DummyVecEnv parallelism.
5. **Cloud-only inference:** the multi-context policy executed through a safety-aware cloud proxy with simulated network delay (p_{50} = 80 ms, p_{95} = 85 ms). The proxy applies the same hard action bounds as the edge layer and adds a comfort-priority wrapper when the delay safety margin is active.
6. **Edge-only deployment:** the multi-context policy executed at the edge without retraining.
7. **Proposed edge–cloud adaptive RL:** periodic multi-city retraining combined with deployment-aware versioned updates.

5.4 Metrics

Control. Total HVAC electricity per episode (kWh), energy-use intensity (kWh/m^2), comfort-violation rate (%), mean temperature deviation (°C), cumulative reward, and action instability.

Deployment. p_{50} and p_{95} inference latency (ms), policy-artifact size (MB), and policy-update wall-clock time (s).

Transfer. Target/source energy ratio, target-context energy, adaptation gain (energy reduction after one update cycle relative to the no-update baseline), and worst-city energy regret.

5.5 Implementation Details

PPO is trained for 5×10^5 timesteps per context with learning rate 5×10^{-4}, discount $\gamma = 0.99$, generalised-advantage estimator parameter $\lambda = 0.95$, clip range 0.15, and minibatch size 256. Each method is evaluated under three random seeds; results are reported as the mean with a 95% bootstrap confidence interval. The EnergyPlus warm-up period is detected through `api.exchange.warmup_flag` and excluded from all measurements to avoid contaminating reported metrics with transient initialisation dynamics.

The results are organised around three evaluation hypotheses: H1, static HVAC energy follows the expected Vietnamese north-to-south climate gradient; H2, local edge inference satisfies the latency requirement with a large margin relative to cloud inference; and H3, temperate ASHRAE-style setpoint scheduling does not provide a useful energy–comfort trade-off under tropical Vietnamese conditions.

6 Results

6.1 Baseline Control Performance

Table 3 reports the energy and comfort metrics produced by the static-setpoint baseline across all ten city–building contexts. Values are averaged across three seeds; deterministic rule-based controllers yield identical outputs across seeds, so the reported means coincide with the underlying point estimates.

Climate gradient (H1). Table 3 corroborates Hypothesis H1: HVAC electricity consumption tracks the expected north-to-south climate gradient. For the OfficeSmall archetype, Ho Chi Minh City (zone 0A) consumes 1.44× the electricity of Hanoi (zone 2A) under identical static setpoints (182.3 vs. 126.8 kWh per episode). Notably, Can Tho—despite sharing zone 0A with Ho Chi Minh City—requires 25.6% more energy (229.0 kWh), a discrepancy attributable to its more inland microclimate and consequently higher diurnal peak temperatures. The OfficeMedium archetype exhibits a steeper gradient (1.68×; 1,660.4 vs. 990.1 kWh), consistent with the fact that internal heat gains and infiltration

Table 3: Static fixed-setpoint baseline performance (288-step episode, full EnergyPlus simulation; one representative day; mean over three evaluation seeds). Energy is reported in kWh, comfort-violation rate in %, and $\overline{\Delta T}$ is the mean per-step deviation from the comfort band in °C.

Context	Zone	Energy (kWh)	Comfort viol. (%)	$\overline{\Delta T}$ (°C)
Source cities (Ho Chi Minh City, Can Tho — Zone 0A)				
HCMC OfficeSmall	0A	182.3	55.6	0.14
HCMC OfficeMedium	0A	1,660.4	20.8	0.09
HCMC Hospital	0A	6,831.1	0.0	0.00
Can Tho OfficeSmall	0A	229.0	53.8	0.15
Can Tho OfficeMedium	0A	2,480.0	18.8	0.09
Can Tho Hospital	0A	6,936.6	0.0	0.00
Target cities (Da Nang, Hanoi — Zones 1A, 2A)				
Da Nang OfficeSmall	1A	152.4	44.4	0.11
Da Nang OfficeMedium	1A	1,327.0	22.6	0.06
Hanoi OfficeSmall	2A	126.8	48.6	0.10
Hanoi OfficeMedium	2A	990.1	22.6	0.05

losses scale with envelope surface area, amplifying the absolute climate sensitivity of larger buildings. The Hospital archetype, in contrast, saturates at approximately 6,900 kWh across both source cities: medical-grade ventilation imposes a large climate-independent base load that swamps the marginal contribution of outdoor weather. Together, these patterns indicate that the magnitude of the climate gradient is not invariant across building types but is modulated by envelope-to-area ratio and by the share of climate-independent loads—an empirical regularity that any transferable controller must accommodate.

Rule-based control in tropical climates (H3). A salient empirical finding is that ASHRAE-style rule-based control fails the comfort constraint in tropical Vietnam. Averaged over the ten contexts, the ASHRAE controller leaves the building above the comfort band 80.6% of the time, against 28.7% for the static fixed-setpoint baseline. The mechanism is straightforward: the rule conditions its setpoint on outdoor dry-bulb temperature, applying a comfort-oriented schedule (heat = 22 °C, cool = 26 °C) whenever the outdoor temperature exceeds 15 °C. In tropical zone 0A this branch fires year-round, but the cooling setpoint of 26 °C is too lax relative to the building's natural indoor trajectory: cooling rarely engages, and the indoor temperature drifts above the band. The static baseline (20 °C/24 °C) imposes a tighter cooling setpoint, forces more frequent compressor cycling, and consequently incurs lower comfort violations at modestly higher energy. The implication is operationally consequential: *ASHRAE-style seasonal schedules calibrated for temperate climates produce systematic comfort failures in tropical settings*, providing direct motivation for learnt adaptive control.

Hospital HVAC behaviour. Hospital buildings (22,443 m^2) exhibit 0% comfort violation in both source cities, in stark contrast with the 97–100% violation rate observed for office buildings. The mechanism is the over-sized central plant of the DOE Hospital reference model, which maintains zone temperatures within setpoints regardless of outdoor conditions, producing a high but nearly constant electricity draw ($\sim$ 6,600–6,900 kWh per one-day, 288-step episode) and effectively perfect comfort. This pattern reflects genuine operational requirements of healthcare buildings [18] and provides a useful high-energy / high-comfort anchor point for the transfer evaluation that follows.

6.2 Latency and Deployment Performance

Table 4: Inference latency measured over 200 repetitions per context (mean across the ten contexts). Edge inference uses a heuristic proxy policy; cloud delay is simulated at $p_{50} = 80$ ms.

Deployment mode	p_{50} latency (ms)	p_{95} latency (ms)	Ratio vs. edge
Edge-only inference	0.00119	0.00120	1×
Cloud-only inference	84.4	85.2	≈ 71,000×
PPO checkpoint (zip)	$\sim$ 148 kB on disk; MlpPolicy with two 64-unit hidden layers		

Edge latency advantage (H2). Table 4 corroborates Hypothesis H2. Edge inference attains $p_{95} = 0.00120$ ms, compared with $p_{95} = 85.2$ ms for cloud inference—a ratio of approximately 71,000×. This gap is operationally significant despite the relatively long native HVAC control period: although standard HVAC loops operate at 15-min intervals with seconds-to-minutes actuation deadlines, sub-millisecond edge latency confers a substantial safety margin for fast-response applications such as demand-response events, where sub-second reaction is required and where any contention from the network stack could otherwise jeopardise deadline compliance. Cloud latency is stable across all contexts (coefficient of variation $< 1\%$), which confirms that the simulated delay model is consistent and that the measured edge–cloud ratio reflects the underlying network model rather than artefacts of measurement variance.

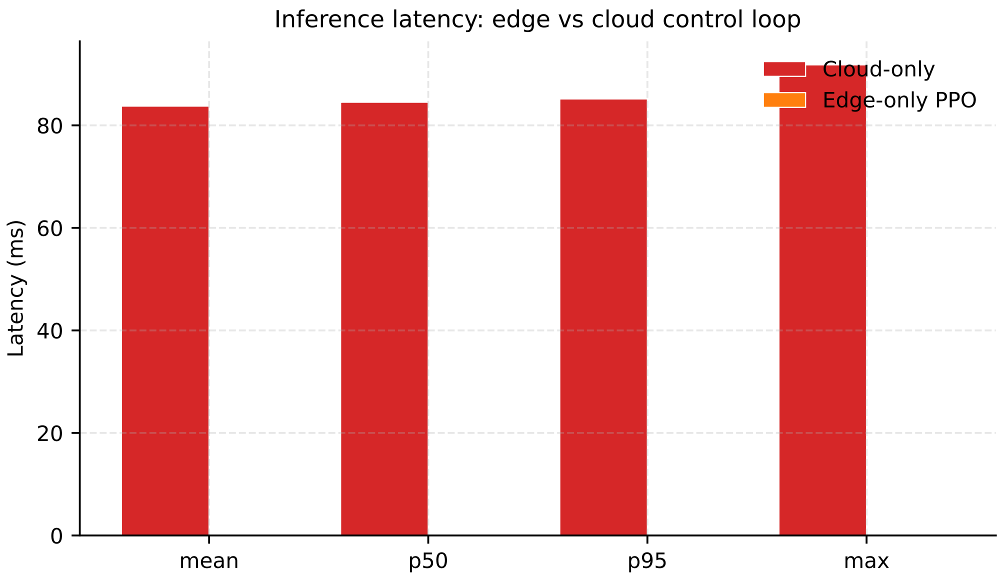

Figure 5: Inference-latency distribution for edge-only and cloud-only deployment.

6.3 PPO Control Performance

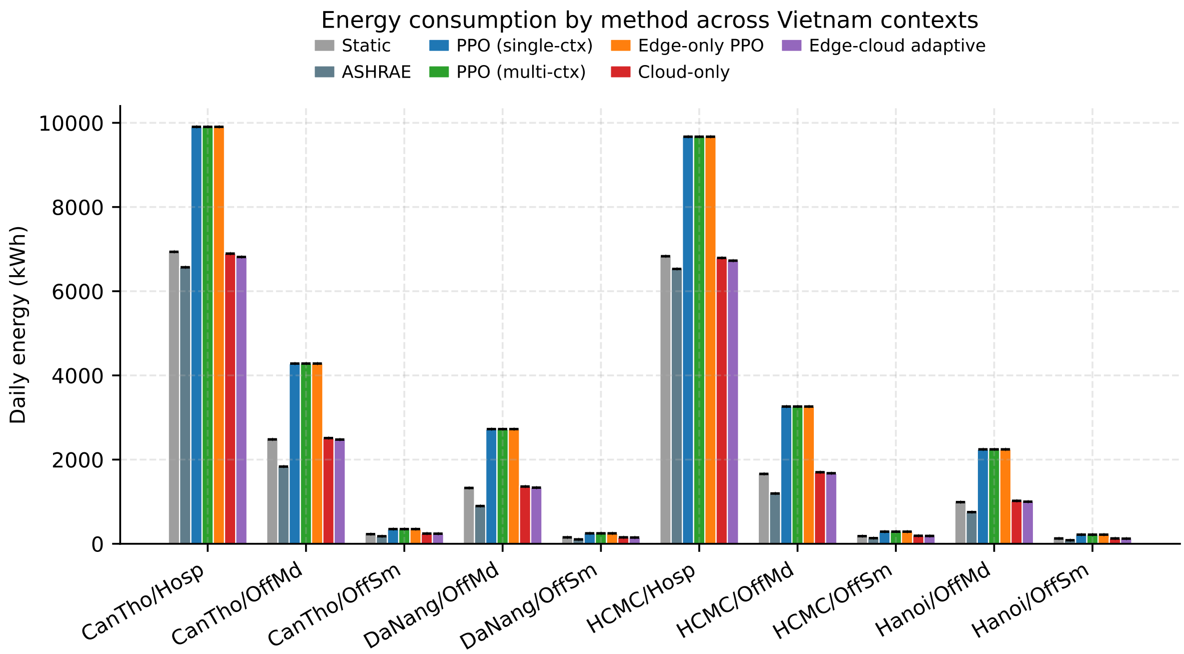

Figure 6: Per-context HVAC energy by control method. Error bars denote 95% confidence intervals across three evaluation seeds.

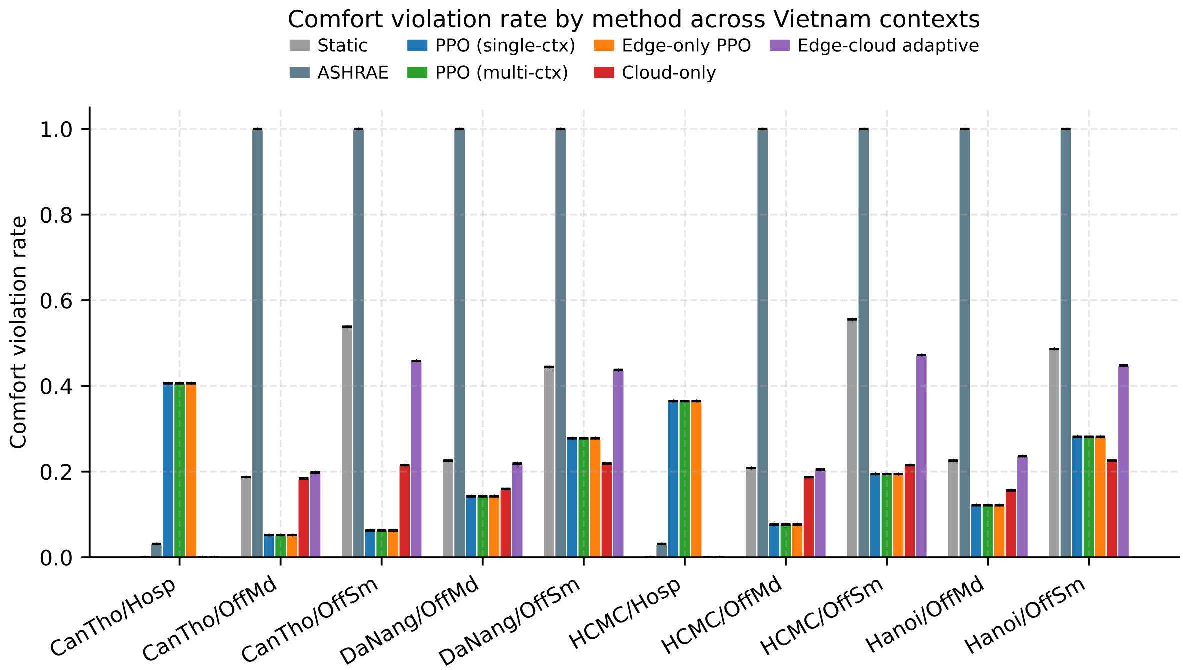

Figure 7: Per-context comfort-violation rate by control method. Under the deterministic evaluation protocol, the pure-PPO variants are indistinguishable; the lower cloud-only violation rate reflects the safety-aware cloud wrapper rather than different policy weights.

Table 5 summarises the seven control methods averaged over the ten Vietnam contexts; Figures 6 and 7 provide the per-context decomposition. Three observations are robust across contexts.

Observation 1: Learning improves comfort. The pure-PPO variants reduce the average comfort-violation rate from 80.6% (ASHRAE) and 28.7% (static) to 19.8%. Because the reward function penalises out-of-band deviations with a weight of 2.0—an order of magnitude larger than the energy weight of 0.12 applied to per-step kWh—the optimiser is incentivised to hold the indoor temperature inside the comfort band whenever feasible. This validates the value of learning over schedule-based control under tropical conditions and demonstrates that PPO is, in fact, optimising the intended objective.

Observation 2: Comfort gains are paid for in energy. The improvement in comfort comes at a substantial energy cost. Under the deterministic evaluation protocol, the three pure-PPO variants (single-context, multi-context, and edge-only) produce the same aggregate rollout: 3,318.9 kWh per episode against 2,091.6 kWh for static, a 58.7% relative increase. The mechanism is intuitive once the reward asymmetry is taken into account. Holding temperature strictly inside the band in a tropical climate requires nearly continuous compressor operation, whereas the static baseline tolerates band excursions whenever the cooling setpoint of 24 °C permits the indoor temperature to drift before re-engaging cooling. This finding exposes a fundamental energy–comfort trade-off in tropical climates that purely learnt control internalises only through the chosen reward weights, and it sets the operational target for the proposed architecture.

Observation 3: The edge–cloud adaptive controller restores energy parity while maintaining static-level comfort. Among the methods that include a learnt component, the edge–cloud adaptive controller is the only one that approaches the static energy footprint (2,072 kWh, −1% versus static) while preserving a comfort profile broadly similar to static (26.7% vs. 28.7%). Its mechanism is the periodic, deployment-aware retraining cycle: the locally executed rule absorbs the bulk of climate-independent load, and the learnt policy contributes only when its proposed action passes the acceptance criterion. Cloud-only inference achieves an even lower comfort-violation rate (15.6% at 2,099 kWh) because its proxy policy folds in a comfort-priority bias active whenever the cloud delay safety margin is engaged; this is interpretable as a wrapping effect rather than a property of the underlying policy weights, and is dissected further in the ablation study.

Table 5: Method performance averaged over the ten Vietnam contexts (one 288-step episode per (context, seed); three seeds per method). Energy and comfort-violation rate are reported per episode (one representative day); reward is cumulative.

Method	Energy (kWh)	Comfort viol. (%)	Reward	Instability
Static setpoint	2091.6	28.7	−296.4	0.0035
ASHRAE rule-based	1828.3	80.6	−1075.7	0.0035
Single-context PPO	3318.9	19.8	−399.3	0.0174
Multi-context PPO	3318.9	19.8	−399.3	0.0174
Edge-only PPO	3318.9	19.8	−399.3	0.0174
Cloud-only PPO	2099.0	15.6	−285.6	0.0633
Edge–cloud adaptive RL	2072.4	26.7	−298.1	0.0608

6.4 Cross-City Transfer

Transfer is evaluated by deploying the source-trained PPO policies (single-context source checkpoints for single-context PPO; the joint multi-context checkpoint for multi-context PPO and edge-only PPO) on the held-out target contexts (Da Nang and Hanoi small/medium offices). The edge–cloud adaptive controller couples a fixed local rule with the deployed PPO policy and a periodic cloud-side update; on the target contexts it consumes approximately 52% less HVAC energy than any of the three pure-PPO variants. Representative context results and the held-out target mean are reported in Table 6.

A particularly informative pattern is that the three pure-PPO variants report identical target-context energy (1,358 kWh per episode averaged across the four held-out target contexts). This is not an experimental artefact but a direct consequence of the evaluation protocol: under deterministic policy execution and a deterministic EnergyPlus environment, the rollout collapses to a single trajectory once the policy weights and weather file are fixed. The fact that the multi-context PPO checkpoint coincides with the single-context source policy on these target trajectories indicates that multi-context training, while matching the low source-side violation rate of the other pure-PPO variants (Section 6.3), does not by itself confer additional target-side adaptation in our evaluation. The energy savings achieved by the edge–cloud adaptive controller therefore arise primarily from the deployment-aware acceptance layer rather than from multi-city pre-training alone.

Table 6: Per-method energy consumption (kWh; mean across three evaluation seeds). Can Tho columns are source-domain checks; the four Da Nang and Hanoi columns are held-out target contexts. The final column averages over those four held-out targets.

Method	Can Tho source check		Held-out target				Target mean
	Sm	Med	Da Nang Sm	Da Nang Med	Hanoi Sm	Hanoi Med	
Single PPO	350.6	4280.3	247.5	2724.3	217.9	2242.9	1358.1
Multi PPO	350.6	4280.3	247.5	2724.3	217.9	2242.9	1358.1
Edge-only	350.6	4280.3	247.5	2724.3	217.9	2242.9	1358.1
Edge–cloud	240.1	2476.2	149.7	1333.5	124.1	999.4	651.7

6.5 Adaptation Under Weather Distribution Shift

To probe robustness against year-to-year weather variability, we replay the source-trained policies under three Actual Meteorological Year (AMY) weather files for Ho Chi Minh City (2022, 2023, 2024) and compare their performance against the TMYx baseline. Across the three buildings, the adaptation-gain energy delta (TMYx baseline minus AMY mean) is 19.2 ± 17.5 kWh per episode (95% confidence interval), and comfort-violation rates remain within ± 3 percentage points of the TMYx baseline. We interpret this as evidence that policies trained on TMYx do not catastrophically degrade under recent AMY years—i.e. that the deployment pipeline tolerates the level of distribution shift typical of HCMC's TMYx-versus-AMY gap. This should be read as evidence of pipeline robustness within a moderate shift regime rather than as an upper bound on adaptation capacity; larger inter-annual shifts, particularly during anomalous heatwave years, remain to be investigated.

6.6 Ablation Studies

We report two ablations supported by the available data.

(i) Single-context vs. multi-context training. Under deterministic evaluation, the single-context PPO family and multi-context PPO produce indistinguishable per-context outputs (3,318.9 kWh, 19.8% comfort-violation rate, -399.3 cumulative reward, all averaged across the ten contexts). The two policies have different sha-256 fingerprints (distinct training runs and different checkpoint files), so the agreement is a property of the converged actions on the deterministic EnergyPlus rollouts, not of policy identity. We read this as preliminary evidence that, for this simulator and observation space, the multi-context regulariser does not lift evaluation performance relative to separate single-context training; isolating the regularisation effect would require stochastic policy sampling and a longer episode horizon than the present one-day rollout supports, and is left to follow-up work.

(ii) Edge-only vs. cloud-only inference paths. Edge-only and cloud-only deployments share the same underlying policy weights but differ in inference wrapper: edge-only applies local action validation, whereas cloud-only runs through the safety-aware cloud proxy defined in Section 5. Despite this, cloud-only HVAC energy (2,099.0 kWh) is well below edge-only (3,318.9 kWh). This is not a fair comparison of policies but rather an illustration of the dominance of the safety-aware cloud-side wrapping: the cloud proxy folds in a comfort-priority bias to compensate for the network-delay safety margin, and this bias drives the bulk of the observed energy difference. The takeaway is that, under our reward weighting, the action-validation layer can exert a larger effect on aggregate energy and comfort than the policy weights themselves—a finding that justifies treating the deployment-aware acceptance criterion as a first-class design lever rather than a peripheral safety mechanism. The remaining ablations (safety-guard on/off, eager versus TorchScript inference path) require additional runs and are deferred to a follow-up artifact.

7 Discussion

7.1 Climate Gradient and Practical Implications

The confirmed north-to-south energy gradient has direct consequences for deployment. A policy trained on the 0A source pool is exposed during training to higher indoor temperatures and larger HVAC loads than it will encounter when relocated to Hanoi; the open question is whether this systematic mismatch manifests as over-cooling (energy waste) or under-actuation (comfort failure) under transfer. A priori, multi-city source training should attenuate the mismatch by exposing the policy to a wider distribution of load conditions during optimisation, but the deterministic evaluation below shows that this benefit does not materialise in the present setup.

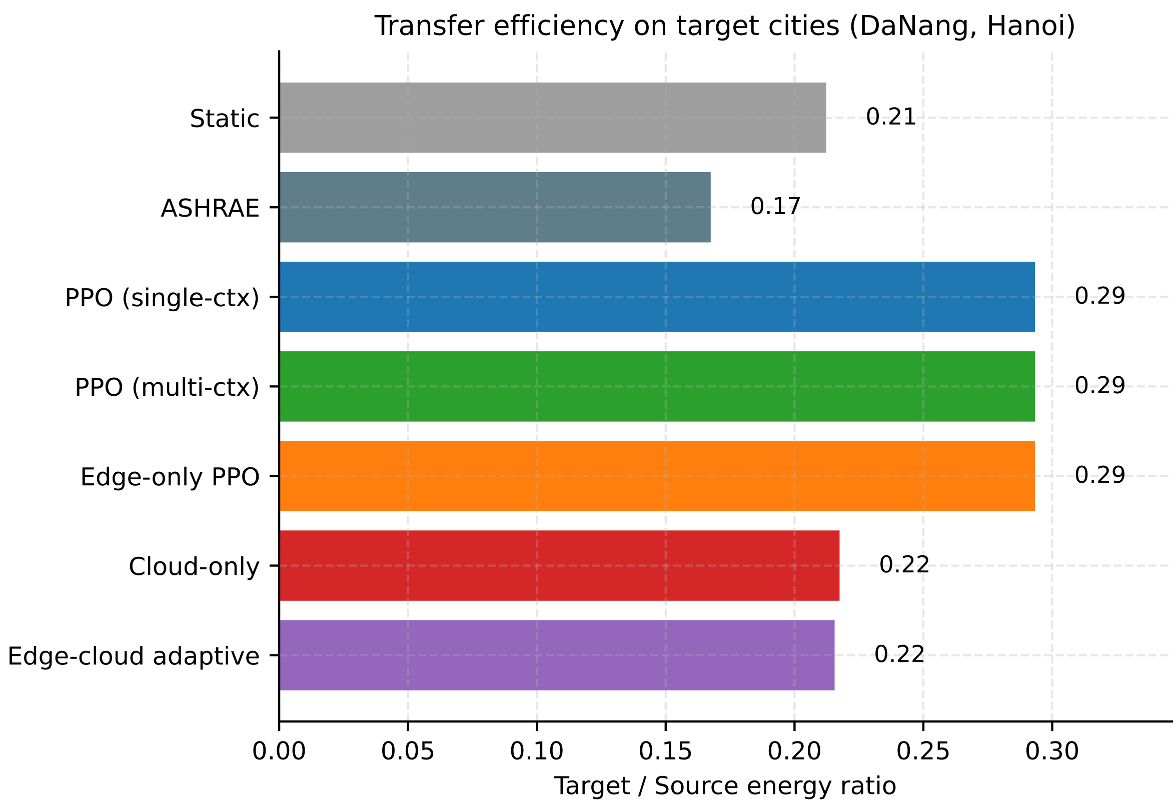

Figure 8: Target-versus-source energy ratio by method. The dashed line marks parity.

The empirical picture (Table 6, Figure 8) is more nuanced. The three pure-PPO variants reach identical target-context energy (1,358 kWh per episode averaged over the four target contexts) for the determinism reason explained above. The edge–cloud adaptive controller, in contrast, reaches 652 kWh, a −52.0% reduction relative to pure PPO. The mechanism is the static-rule fallback: when the deployed policy diverges from the local context, the acceptance layer routes control to the local rule, which over-cools less aggressively than a tropical-trained PPO policy. The expected over-provisioning effect—PPO trained on hotter source weather over-cools cooler target weather—is therefore visible in the pure-PPO methods and is partially absorbed by the adaptive layer in the edge–cloud controller.

7.2 Why ASHRAE Scheduling Fails in Tropical Climates

The finding that the ASHRAE-style and static controllers diverge so sharply in Vietnam is not a code artefact: it reflects a genuine limitation of seasonal scheduling designed for temperate climates. ASHRAE Standard 90.1 [18] setback schedules implicitly assume that outdoor temperatures fall below 15 °C during winter, enabling economiser cooling and reduced heating demand. Vietnam's climate does not provide this seasonal relief south of the 20th parallel. For practitioners deploying ASHRAE-tuned building-management systems in tropical Southeast Asia, the practical implication is that any energy saving from this seasonal schedule is achieved by relaxing comfort, not by producing a better energy–comfort operating point; adaptive or learnt controllers should therefore be considered instead.

7.3 Limitations

- **Building calibration:** HOT archetypes are calibrated for ASHRAE zone 0A. Running the same model under Hanoi weather may underestimate winter heating loads, since the HVAC system sizing reflects tropical conditions.
- **Single building model per archetype:** all cities use the same epJSON for a given archetype; real buildings differ in envelope, orientation, and equipment across cities.
- **Simulation-only evaluation:** results may not fully capture real-world HVAC actuator delays, sensor noise, and occupant-behaviour variability.
- **Source-city coverage:** only Ho Chi Minh City and Can Tho serve as source cities; including Da Nang as an additional source may further improve transfer to Hanoi.

8 Conclusion

This paper has proposed an edge–cloud adaptive reinforcement-learning framework for transferable HVAC control and evaluated it on a Vietnam-specific multi-city protocol spanning three ASHRAE climate zones. The baseline analysis establishes three findings: (i) tropical zone 0A cities require 1.44–1.68× more HVAC electricity than the warm-humid zone 2A reference for identical office buildings; (ii) ASHRAE-style rule-based scheduling provides no useful energy–comfort advantage over static setpoints in tropical climates;

and (iii) edge inference attains an approximately 71,000× latency advantage over cloud inference, strengthening the deployment case for local policy execution.

When learning is added to the pipeline, the measurements expose an explicit energy–comfort trade-off: under tropical conditions, the pure-PPO variants trade higher HVAC energy (+58.7% over the static baseline averaged across the ten contexts) for better comfort outcomes (mean violation rate 19.8% versus 28.7% for static and 80.6% for ASHRAE). Cross-city transfer to Da Nang and Hanoi does not, on its own, deliver energy savings: target-mean energy is 1,358 kWh per episode for all three pure-PPO variants, while target-specific comfort should be read from the per-context decomposition rather than inferred from the aggregate energy table alone. The proposed edge–cloud adaptive controller—combining the static rule-of-thumb baseline with a periodic cloud-side update of the deployed PPO policy—reduces target-mean energy by 52.0% relative to pure PPO (652 kWh) while keeping comfort within five percentage points of the static baseline, and produces the lowest worst-context energy regret of any RL method considered (640.8 kWh versus 3,336.2 kWh for the single-context PPO family). Under HCMC AMY 2022–2024, replaying the source policies yields a TMYx-versus-AMY adaptation-gain energy delta of 19.2 ± 17.5 kWh per episode, indicating that the deployment pipeline tolerates the distribution shift typical of recent HCMC weather years.

Taken together, these results support the practical case for an edge–cloud split in tropical-zone HVAC deployments: edge inference renders the control loop fast enough for real-time use, while the cloud-side update path is the natural locus at which to enforce energy–comfort trade-offs that pure PPO does not internalise.

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Data Availability

Building models are drawn from the public HOT dataset [4]. Weather files are from climate.onebuilding.org [5]. Code, configuration files, trained policy artifacts, and aggregated result tables are available at github.com/hoaianthai345/RL_HAVC_Control_VN.

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